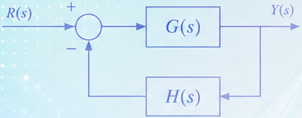

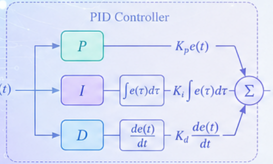

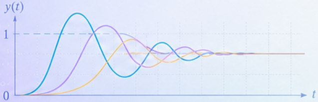

Stability

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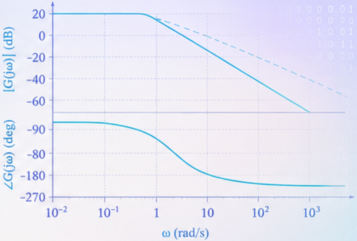
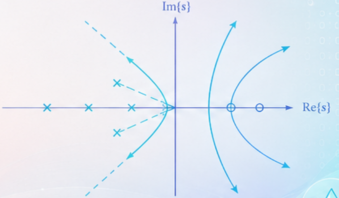
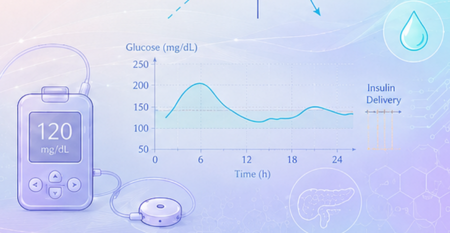




$$G(s) = \frac{Y(s)}{U(s)} \qquad a_n \frac{d^n y(t)}{dt^n} + \cdots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \cdots + b_m \frac{d^m u(t)}{dt^m} + \cdots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$



Stability: Basic Idea

The total response of an LTI system is

$$\boxed{c(t) = c_{\text{forced}}(t) + c_{\text{natural}}(t)}$$

Forced Response

Produced by the external input.

$$c_{\text{forced}}(t)$$

It usually determines the desired steady-state behaviour.

Natural Response

Produced by system dynamics and initial conditions.

$$c_{\text{natural}}(t)$$

Its behaviour determines stability.

For a stable system,

$$\boxed{c_{\text{natural}}(t) \to 0 \qquad \text{as } t \to \infty}$$

Stable, Unstable, and Marginally Stable Systems

Stable

$$c_n(t) \to 0$$

The natural response decays with time.

$$e^{-at}, \qquad a > 0$$

Unstable

$$|c_n(t)| \to \infty$$

The natural response grows without bound.

$$e^{at}, \qquad a > 0$$

Marginally Stable

The natural response neither decays nor grows.

$$\sin(\omega t), \qquad \cos(\omega t)$$

It remains bounded and oscillatory.

BIBO Stability

BIBO means

$$\boxed{\text{Bounded Input} \Rightarrow \text{Bounded Output}}$$

Stable

If every bounded input satisfies

$$|u(t)| \leq M_u < \infty,$$

and produces

$$|y(t)| \leq M_y < \infty,$$

then the system is BIBO stable.

Unstable

If there exists a bounded input for which

$$|u(t)| \leq M_u,$$

but

$$|y(t)| \to \infty,$$

then the system is unstable.

Closed-Loop Poles and Stability

For an LTI system, stability can be determined from the locations of its closed-loop poles.

Stable System

All poles lie in the open left-half plane:

$$\boxed{\mathrm{Re}(p_i) < 0}$$

Typical natural responses are

$$e^{-at}, \qquad e^{-at}\sin(\omega t).$$

Both decay to zero.

Unstable System

At least one pole lies in the right-half plane:

$$\boxed{\mathrm{Re}(p_i) > 0}$$

Typical natural responses are

$$e^{at}, \qquad e^{at}\sin(\omega t).$$

Their magnitude increases with time.

Imaginary-Axis Poles

Simple Imaginary-Axis Poles

For poles

$$p_{1,2} = \pm j\omega,$$

the natural response contains

$$\sin(\omega t), \qquad \cos(\omega t).$$

The amplitude remains constant.

$$\Rightarrow \quad \textit{MarginallyStable}$$

Repeated Imaginary-Axis Poles

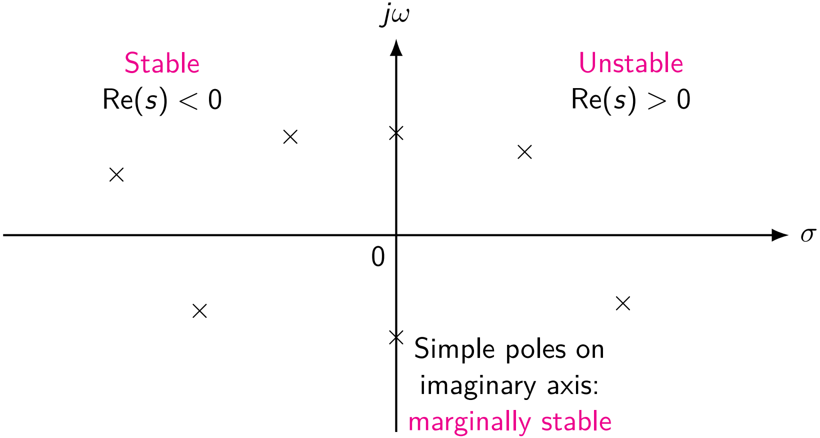
Repeated poles generate terms such as

$$t \sin(\omega t), \qquad t \cos(\omega t).$$

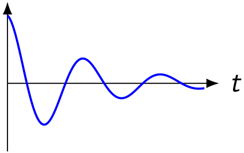
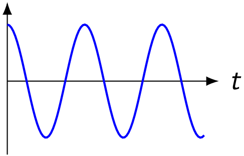
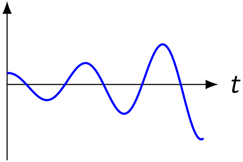
Their amplitude increases with time.

$$\Rightarrow \quad \textit{Unstable}$$

Stability Regions in the s-Plane



Natural Response and Stability

Stable	Marginally Stable	Unstable
$p = -a, \qquad a > 0$	$p = \pm j\omega$	$p = +a, \qquad a > 0$
		
$c_n(t) \to 0$	$\lvert c_n(t) \rvert < \infty$	$\lvert c_n(t) \rvert \to \infty$

Example: Stable Closed-Loop System

For the unity-feedback system

$$G(s) = \frac{3}{s(s+1)(s+2)},$$

the closed-loop transfer function is

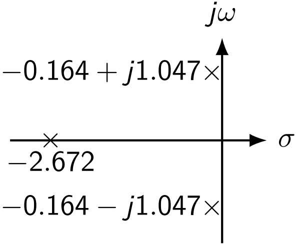
$$T(s) = \frac{3}{s^3 + 3s^2 + 2s + 3}.$$

The closed-loop poles are approximately

$$p_1 = -2.672, \quad p_{2,3} = -0.164 \pm j1.047.$$

Since [Re(p_i) < 0, all poles lie in the left-half plane.

$\boxed{\text{System is stable}}$



Example: Unstable Closed-Loop System

For the unity-feedback system

$$G(s) = \frac{7}{s(s+1)(s+2)},$$

the closed-loop transfer function is

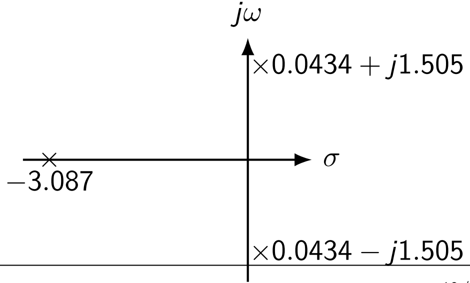
$$T(s) = \frac{7}{s^3 + 3s^2 + 2s + 7}.$$

The closed-loop poles are approximately

$$p_1 = -3.087, \quad p_{2,3} = 0.0434 \pm j1.505.$$

Since the complex poles satisfy $\mathrm{Re}(p_{2,3}) > 0$, they lie in the right-half plane.

$\boxed{\text{System is unstable}}$



Stability Summary

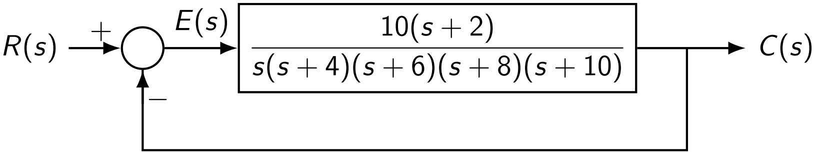
Pole Location	Natural Response	Stability
$\mathrm{Re}(p) < 0$	Decays to zero	Stable
$\mathrm{Re}(p) > 0$	Grows without bound	Unstable
Simple $p = \pm j\omega$	Sustained oscillation	Marginally stable
Repeated $p = \pm j\omega$	Growing oscillation	Unstable

Important

For a rational continuous-time LTI system to be BIBO stable, all poles must lie strictly in the open left-half plane.

Determining Stability of a Feedback System

It is not always straightforward to determine whether a feedback control system is stable.



- The poles of the forward transfer function $G(s)$ are easy to identify.
- However, stability depends on the closed-loop poles.
- These poles are obtained from the closed-loop characteristic equation

$$\boxed{1 + G(s) = 0}.$$

Equivalent Closed-Loop System

For unity negative feedback,

$$T(s) = \frac{G(s)}{1 + G(s)}.$$

For

$$G(s) = \frac{10(s+2)}{s(s+4)(s+6)(s+8)(s+10)},$$

we obtain

$$T(s) = \frac{10(s+2)}{s(s+4)(s+6)(s+8)(s+10) + 10(s+2)}.$$

Equivalent Closed-Loop System

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$$T(s) = \frac{10(s+2)}{s(s+4)(s+6)(s+8)(s+10) + 10(s+2)}.$$

Expanding the denominator,

$$\boxed{T(s) = \frac{10(s+2)}{s^5 + 28s^4 + 284s^3 + 1232s^2 + 1930s + 20}}$$

$$R(s) \rightarrow \boxed{\frac{10(s+2)}{s^5 + 28s^4 + 284s^3 + 1232s^2 + 1930s + 20}} \rightarrow C(s)$$

Why Direct Pole Calculation Becomes Difficult

The closed-loop poles are the roots of

$$\boxed{s^5 + 28s^4 + 284s^3 + 1232s^2 + 1930s + 20 = 0}$$

- This is a fifth-order characteristic equation.
- Factoring such a polynomial by hand may be difficult.
- Numerical software can calculate the roots, but we would like a method that can determine stability without explicitly solving for every pole.

Question

Can we obtain useful information about stability directly from the coefficients of the characteristic polynomial?

A Preliminary Test from Polynomial Coefficients

Consider the characteristic polynomial

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0.$$

For a Stable System

If all poles lie in the left-half plane, the denominator can be formed from factors such as

$$(s + a_i), \qquad a_i > 0,$$

and second-order factors associated with complex conjugate poles having negative real parts.

Therefore, the resulting characteristic polynomial has coefficients with the same sign.

Necessary Conditions

For asymptotic stability,

- all coefficients must have the same sign;
- no power of s should be missing.

If either condition is violated, the system is

$$\boxed{\text{unstable or, at best, marginally stable}}$$

Are Positive Coefficients Sufficient?

Consider again

$$P(s) = s^5 + 28s^4 + 284s^3 + 1232s^2 + 1930s + 20.$$

All coefficients are positive:

$$1, \quad 28, \quad 284, \quad 1232, \quad 1930, \quad 20 > 0.$$

Also, no power of s is missing.

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$$P(s) = s^5 + 28s^4 + 284s^3 + 1232s^2 + 1930s + 20.$$

All coefficients are positive:

$$1, \quad 28, \quad 284, \quad 1232, \quad 1930, \quad 20 > 0.$$

Also, no power of s is missing.

Can we conclude that the system is stable?

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NO

Positive coefficients and no missing powers are necessary conditions, but they are not sufficient to guarantee that all poles lie in the left-half plane.

Are Positive Coefficients Sufficient?

Consider again

$$P(s) = s^5 + 28s^4 + 284s^3 + 1232s^2 + 1930s + 20.$$

All coefficients are positive:

$$1, \quad 28, \quad 284, \quad 1232, \quad 1930, \quad 20 > 0.$$

Also, no power of s is missing.

Can we conclude that the system is stable?

NO

Positive coefficients and no missing powers are necessary conditions, but they are not sufficient to guarantee that all poles lie in the left-half plane.

Therefore, we need a systematic method that determines

the number of characteristic roots in the right-half plane

without explicitly calculating all roots.

Motivation for the Routh–Hurwitz Criterion

For a characteristic polynomial

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0,$$

we would like to determine stability using only its coefficients.

- Direct root calculation may become difficult for high-order systems.
- Inspection of coefficient signs provides only a necessary condition.
- We need to determine whether any roots lie in the right-half s-plane.
- This can be accomplished without explicitly finding the roots.

Next Topic

Routh–Hurwitz Stability Criterion

Routh–Hurwitz Stability Criterion

Consider the characteristic polynomial of a closed-loop system:

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0.$$

- The system is stable if all roots of $P(s) = 0$ lie in the left-half plane.
- Explicitly calculating the roots may be difficult for a high-order polynomial.
- The Routh–Hurwitz criterion determines stability directly from the coefficients of $P(s)$.

Main Result

The number of sign changes in the first column of the Routh array equals the number of roots of $P(s)$ in the right-half plane.

Objective of the Routh Criterion

For

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0,$$

the Routh criterion answers:

$$\boxed{\text{How many roots lie in the right-half } s\text{-plane?}}$$

Stable

No roots in the right-half plane:

$$N_{\mathrm{RHP}} = 0.$$

Hence, there are no sign changes in the first column.

Unstable

If the first column contains sign changes,

$$N_{\mathrm{RHP}} > 0,$$

the system has right-half-plane poles and is unstable.

Construction of the Routh Array

Consider

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \cdots + a_1 s + a_0.$$

The first two rows are formed directly from the coefficients:

$$\begin{array}{c|cccc}
s^n & a_n & a_{n-2} & a_{n-4} & \cdots \\
s^{n-1} & a_{n-1} & a_{n-3} & a_{n-5} & \cdots \\
s^{n-2} & b_1 & b_2 & b_3 & \cdots \\
s^{n-3} & c_1 & c_2 & c_3 & \cdots \\
\vdots & \vdots & \vdots & \vdots & \\
s^0 & \cdots & & &
\end{array}$$

Important

The coefficients are arranged in descending powers of s, alternating between the first and second rows.

Calculating the Routh Array Elements

The elements of the third row are obtained from the first two rows.

$$b_1 = \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}}$$

$$b_2 = \frac{a_{n-1}a_{n-4} - a_n a_{n-5}}{a_{n-1}}$$

Similarly,

$$c_1 = \frac{b_1 a_{n-3} - a_{n-1} b_2}{b_1}.$$

Procedure

Continue the same operation row by row until the s^0 row is obtained.

Stability Condition from the Routh Array

After constructing the Routh array, inspect only its first column.

$$\begin{array}{c|c} s^n & r_1 \\ s^{n-1} & r_2 \\ s^{n-2} & r_3 \\ s^{n-3} & r_4 \\ \vdots & \vdots \\ s^0 & r_{n+1} \end{array}$$

- The number of sign changes in

$$r_1,\ r_2,\ r_3, \ldots, r_{n+1}$$

  equals the number of right-half-plane roots.
- For an asymptotically stable system, all entries in the first column must have the same sign.

$$\boxed{\text{No sign change} \quad \Rightarrow \quad \text{no RHP poles}}$$

Example 1: Stable System

Consider

$$P(s) = s^3 + 2s^2 + 3s + 4.$$

The Routh array is

$$\begin{array}{c|cc} s^3 & 1 & 3 \\ s^2 & 2 & 4 \\ s^1 & \dfrac{2(3)-1(4)}{2} & 0 \\ s^0 & 4 & \end{array} \implies \begin{array}{c|cc} s^3 & 1 & 3 \\ s^2 & 2 & 4 \\ s^1 & 1 & 0 \\ s^0 & 4 & \end{array}$$

First Column

$$1, \quad 2, \quad 1, \quad 4$$

There is no sign change.

$$\boxed{\text{System is stable}}$$

Example 2: Unstable System

Consider

$$P(s) = s^3 + 2s^2 + s + 4.$$

The Routh array is

$$\begin{array}{c|cc} s^3 & 1 & 1 \\ s^2 & 2 & 4 \\ s^1 & \dfrac{2(1)-1(4)}{2} & 0 \\ s^0 & 4 & \end{array} \implies \begin{array}{c|cc} s^3 & 1 & 1 \\ s^2 & 2 & 4 \\ s^1 & -1 & 0 \\ s^0 & 4 & \end{array}$$

The first column is

$$1, \qquad 2, \qquad -1, \qquad 4.$$

There are two sign changes:

$$+ \to - \qquad \text{and} \qquad - \to +.$$

$$\boxed{\text{Two RHP poles} \quad \Rightarrow \quad \text{system is unstable}}$$

Stability Range of a Parameter

Consider

$$P(s) = s^3 + 3s^2 + (K+2)s + 4K.$$

The Routh array is

$$\begin{array}{c|cc} s^3 & 1 & K+2 \\ s^2 & 3 & 4K \\ s^1 & \dfrac{3(K+2)-4K}{3} & 0 \\ s^0 & 4K & \end{array}$$

Therefore,

$$s^1: \qquad \frac{6-K}{3}.$$

For stability, every first-column entry must be positive:

$$1 > 0, \qquad 3 > 0, \qquad \frac{6-K}{3} > 0, \qquad 4K > 0.$$

Hence, $0 < K < 6$,

Special Case I: Zero in the First Column

During construction of the Routh array, the first element of a row may become zero while other elements in that row are nonzero. For example,

$$\begin{array}{c|ccc} s^n & \cdots & \cdots & \cdots \\ s^{n-1} & 0 & a & b \end{array}$$

Direct calculation of the next row is then impossible because division by zero occurs.

Remedy: ε-Method

Replace the zero by a small positive number

$$0 \to \varepsilon, \qquad \varepsilon > 0,$$

complete the Routh array, and finally examine the signs as

$$\varepsilon \to 0^+.$$

Special Case II: Entire Row of Zeros

Another special case occurs when an entire row becomes zero:

$$\begin{array}{c|ccc} s^{k+1} & a & b & c \\ s^{k} & 0 & 0 & 0 \end{array}$$

- Form an auxiliary polynomial using the coefficients of the row immediately above the zero row.
- Suppose
$$A(s) = as^{k+1} + bs^{k-1} + cs^{k-3} + \cdots .$$
- Differentiate it:
$$\frac{dA(s)}{ds}.$$
- Replace the zero row by the coefficients of $\dfrac{dA(s)}{ds}$, then continue the Routh array normally.

Meaning of an Entire Row of Zeros

An entire row of zeros indicates that the characteristic polynomial contains roots that are symmetrically located about the origin.
Possible symmetric roots include

$$\pm a, \qquad \pm j\omega, \qquad \pm\sigma \pm j\omega.$$

Important

A row of zeros does not by itself imply stability.
The auxiliary polynomial must be examined to determine the locations of the associated roots.

For simple imaginary-axis roots,

$$s = \pm j\omega,$$

the system may exhibit sustained oscillations and be marginally stable, provided all remaining poles lie in the left-half plane.

Poles of a State-Space System

Consider the LTI state-space model

$$\dot{\boldsymbol{x}}(t) = A\boldsymbol{x}(t) + B\boldsymbol{u}(t), \qquad \boldsymbol{y}(t) = C\boldsymbol{x}(t) + D\boldsymbol{u}(t).$$

The poles of the system are determined by the state matrix A.
They are the values of s satisfying

$$\boxed{\det(sI - A) = 0}$$

This equation is called the characteristic equation.

Equivalently, the poles are the

$$\boxed{\text{eigenvalues of } A}$$

because eigenvalues satisfy

$$\det(\lambda I - A) = 0.$$

Hence,

$$\boxed{p_i = \lambda_i(A)}$$

Example: Finding Poles from the State Matrix

Consider

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}.$$

Form

$$sI - A = \begin{bmatrix} s & -1 \\ 6 & s+5 \end{bmatrix}.$$

The characteristic equation is

$$\det(sI - A) = 0.$$

Therefore,

$$\begin{vmatrix} s & -1 \\ 6 & s+5 \end{vmatrix} = 0.$$

Expanding the determinant,

$$s(s+5) + 6 = 0.$$

Thus,

$$s^2 + 5s + 6 = 0.$$

Factorizing,

$$(s+2)(s+3) = 0.$$

Hence,

$$\boxed{p_1 = -2, \qquad p_2 = -3}$$

Both poles lie in the left-half plane.

State-Space Poles and Transfer-Function Poles

For the state-space model,

$$G(s) = C(sI - A)^{-1}B + D.$$

Using

$$(sI - A)^{-1} = \frac{\mathrm{adj}(sI - A)}{\det(sI - A)},$$

we obtain

$$G(s) = C\frac{\mathrm{adj}(sI - A)}{\det(sI - A)}B + D.$$

Key Connection

The roots of

$$\boxed{\det(sI - A) = 0}$$

are the eigenvalues of A and determine the internal system modes.
For a minimal realization, these are also the poles of $G(s)$.

State-Space Poles and Stability

For

$$\dot{\boldsymbol{x}} = A\boldsymbol{x},$$

the natural response is governed by

$$\boldsymbol{x}(t) = e^{At}\boldsymbol{x}(0).$$

Therefore, the eigenvalues of A determine the behaviour of the natural response.

Stable

$$\mathrm{Re}(\lambda_i) < 0$$

for all i.

Natural response decays.

Unstable

At least one

$$\mathrm{Re}(\lambda_i) > 0.$$

Natural response grows.

Marginal Case

Simple eigenvalues on the imaginary axis:

$$\mathrm{Re}(\lambda_i) = 0.$$

Sustained oscillations may occur.

Thank You

Questions and Discussion